

AEIA Engineering Analysis Report

Dataset: engine_test_run.csv
Rows: 60 | Columns: 7
Session ID: S00001
Generated: 2026-07-12 13:51:01
AEIA Version: 0.1.0

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Statistical Summary

Column	Mean	Std Dev	Min	Max	Q1	Q3	Trend
Engine_Temp_C	91.7507	5.9313	86.8500	134.5000	89.7350	92.7525	Increasing
Oil_Pressure_psi	39.6514	1.6187	36.1200	43.1400	38.5375	40.8725	Stable
RPM	2495.3433	19.2550	2456.8000	2540.0000	2484.7000	2511.2500	Stable
Vibration_mm_s	0.2557	0.0565	0.1860	0.6200	0.2288	0.2690	Stable

Engineering Findings

Engine_Temp_C

[Warning] Column 'Engine_Temp_C' shows an unusual value of 134.5 at row 44, which is 7.21 standard deviations from the mean — flagged as a statistical outlier (Warning).

[Warning] Column 'Engine_Temp_C' has a value of 134.5 at row 44 that falls outside the expected range (below 85.21 or above 97.28) — flagged as an outlier (Warning).

[Info] Column 'Engine_Temp_C' shows a Increasing trend over the dataset (slope = 0.1383).

[Critical] Engine temperature exceeded the critical safety threshold of 120°C. (triggered because Engine_Temp_C = 134.5 at row 44 > 120).

[Warning] This channel shows a sustained upward drift over the recorded samples. (triggered because trend_slope(Engine_Temp_C) = 0.1383 > 0.05).

Vibration_mm_s

[Warning] Column 'Vibration_mm_s' shows an unusual value of 0.62 at row 29, which is 6.45 standard deviations from the mean — flagged as a statistical outlier (Warning).

[Warning] Column 'Vibration_mm_s' has a value of 0.62 at row 29 that falls outside the expected range (below 0.17 or above 0.33) — flagged as an outlier (Warning).

[Warning] This channel shows unusually high variability relative to its average value, indicating possible instability. (triggered because coefficient_of_variation(Vibration_mm_s) = 0.2209 > 0.15).

Engine_Temp_C,Oil_Pressure_psi,RPM,Vibration_mm_s

[Warning] Row 29 was flagged as an unusual combination of values across Engine_Temp_C, Oil_Pressure_psi, RPM, Vibration_mm_s — this pattern does not resemble the rest of the dataset (Warning).

Oil_Pressure_psi,RPM

[Info] Columns 'Oil_Pressure_psi' and 'RPM' are strongly correlated ($r = 0.9336$), suggesting a possible relationship worth investigating.

[Info] Two channels are strongly correlated, which may indicate a shared physical cause or sensor dependency. (triggered because $|r|$ between Oil_Pressure_psi and RPM = 0.9336 > 0.7).

*

[Warning] This row's combination of values across multiple channels does not resemble normal operation, even though no single value looks extreme. (triggered because Row 29 flagged by Isolation Forest).

Conclusion

Analysis identified 12 finding(s) across the dataset. 1 Critical finding(s) require immediate attention. 8 Warning-level finding(s) were detected. 3 informational finding(s) noted. Finding types: correlation pattern, rule match, statistical anomaly, trend pattern.

Recommendations

1. [Critical] Critical: Immediately inspect the cooling system and sensor calibration for the affected channel. (Source: Rule RULE-001: Critical Threshold Breach - Engine Temperature).
2. [Warning] Warning: Schedule a calibration check to rule out sensor drift before the trend approaches operational limits. (Source: Rule RULE-002: Sustained Upward Drift).
3. [Warning] Warning: Cross-check this channel against related sensors for a common root cause. (Source: Rule RULE-003: High Variance Instability).

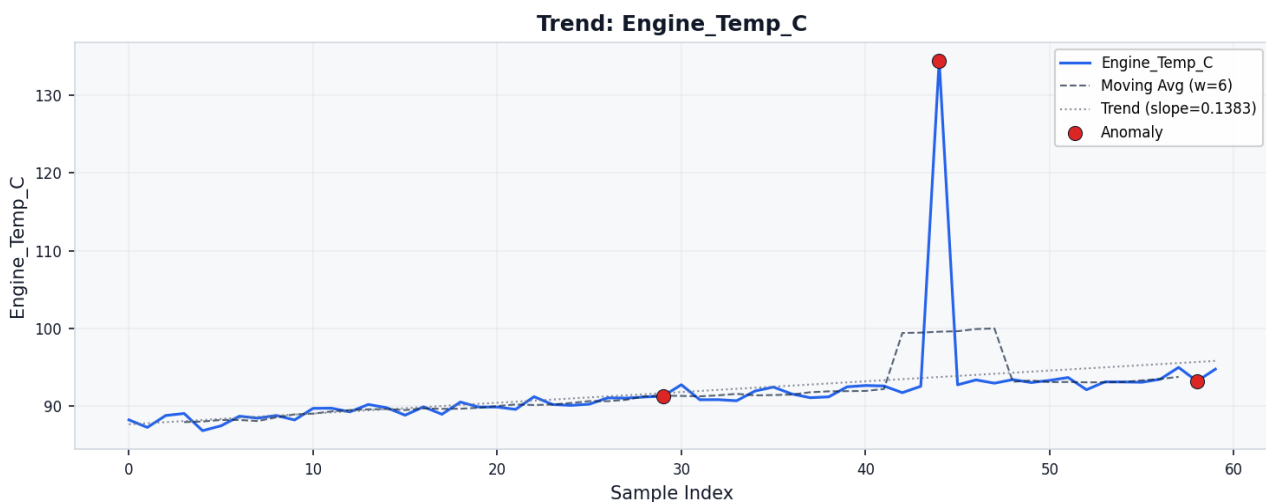
sign-off by a qualified engineer before any operational decision is made.

This report was generated with AI-assisted analysis (AEIA). Findings and recommendations are advisory and require review and

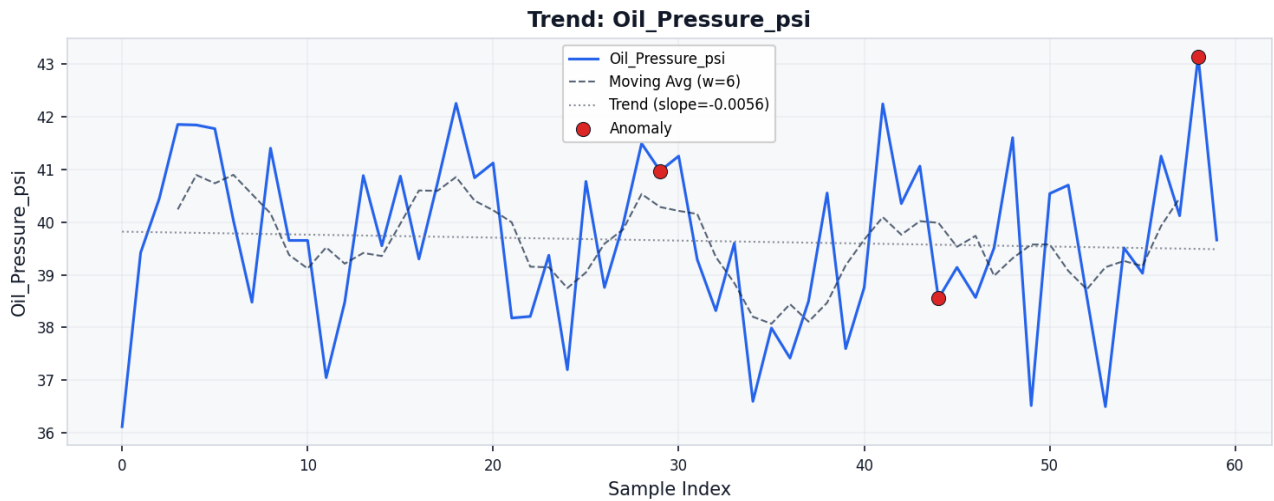
4. **[Warning]** Warning: Review this specific timestamp/sample manually — combined-channel anomalies often indicate sensor cross-talk or a transient event. (Source: Rule RULE-006: Multivariate Anomaly Cluster).
5. **[Warning]** Warning: Review the unusual value(s) in column 'Engine_Temp_C' to determine whether they represent a genuine anomaly or a measurement error. (Source: ZScore detection on column 'Engine_Temp_C').
6. **[Warning]** Warning: Review the unusual value(s) in column 'Vibration_mm_s' to determine whether they represent a genuine anomaly or a measurement error. (Source: ZScore detection on column 'Vibration_mm_s').
7. **[Warning]** Warning: Investigate the out-of-range value(s) in column 'Engine_Temp_C' to assess whether they indicate a system issue. (Source: IQR detection on column 'Engine_Temp_C').
8. **[Warning]** Warning: Investigate the out-of-range value(s) in column 'Vibration_mm_s' to assess whether they indicate a system issue. (Source: IQR detection on column 'Vibration_mm_s').
9. **[Warning]** Warning: Examine the flagged rows for unusual multivariate patterns. These may indicate a system-level anomaly not visible in individual columns. (Source: IsolationForest detection on column 'Engine_Temp_C,Oil_Pressure_psi,RPM,Vibration_mm_s').
10. **[Info]** Info: Consider whether one channel could serve as an early-warning proxy for the other. (Source: Rule RULE-005: Strong Correlated Channels).

Graphs

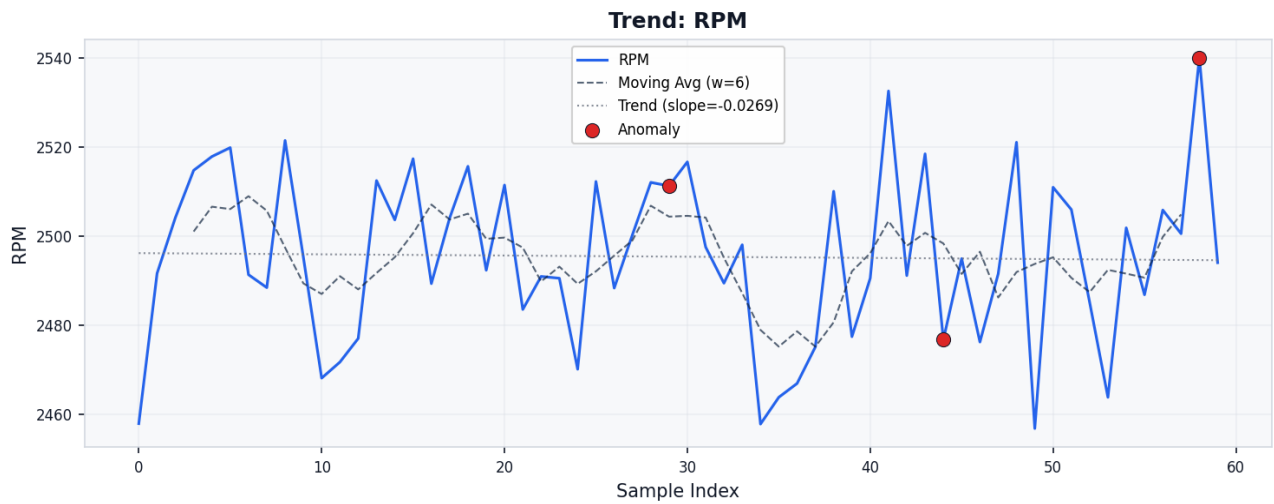
Trend Engine Temp C



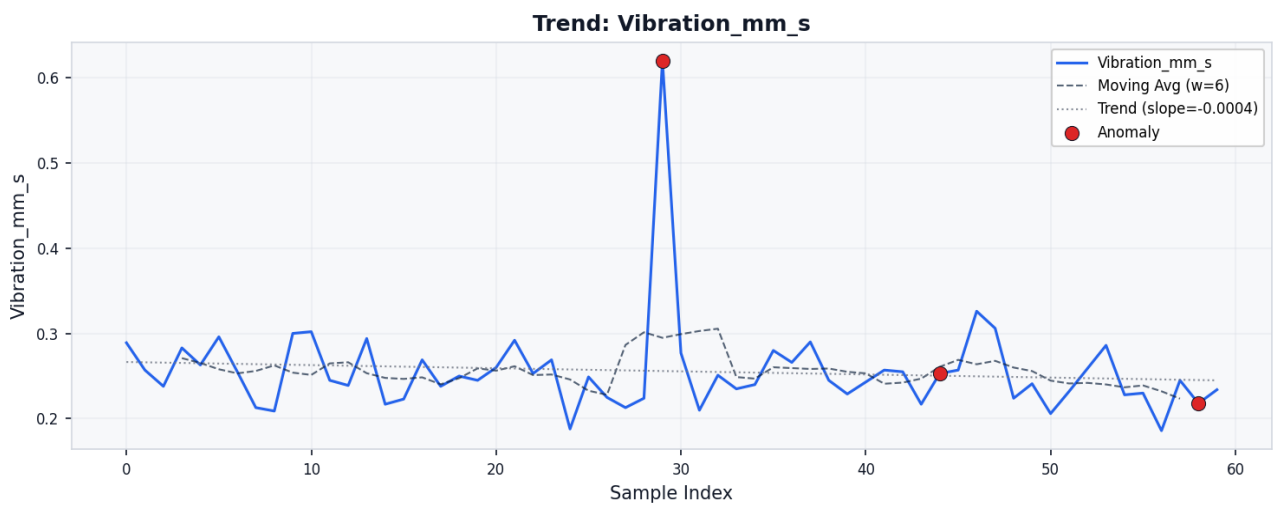
Trend Oil Pressure Psi



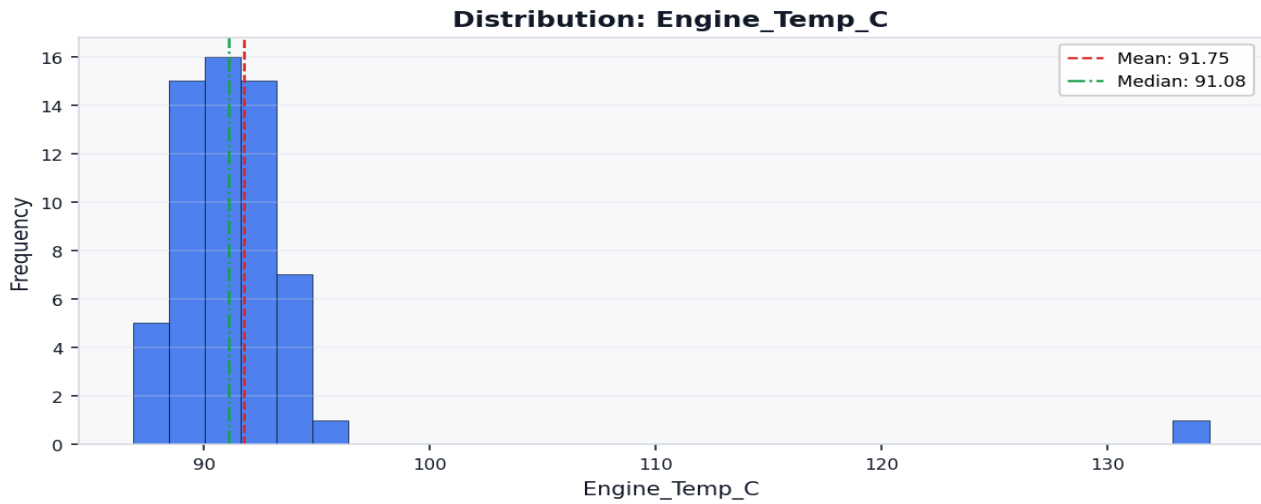
Trend Rpm



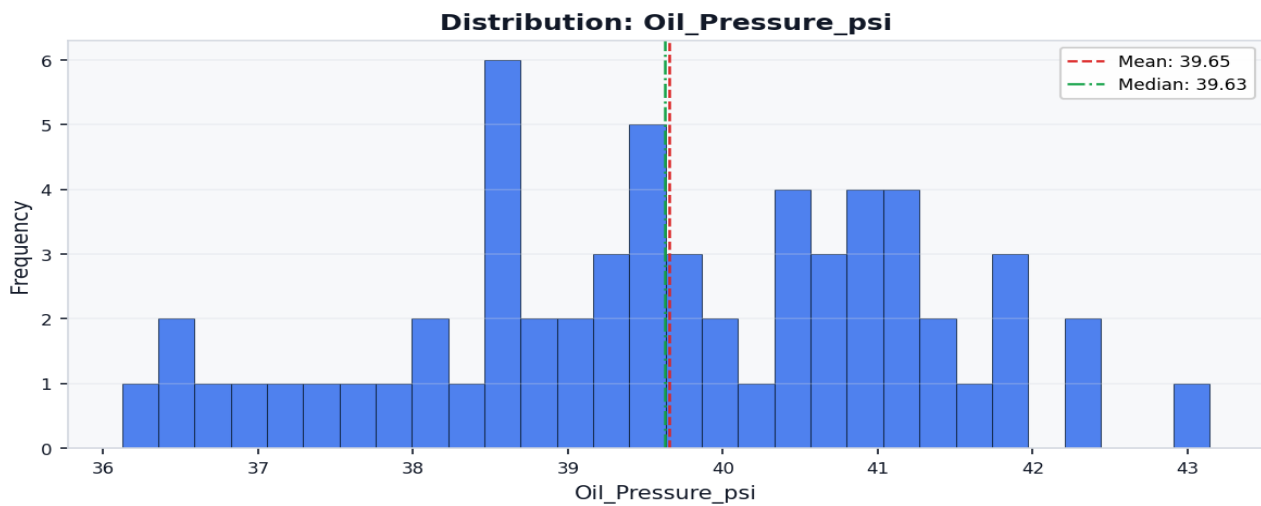
Trend Vibration Mm S



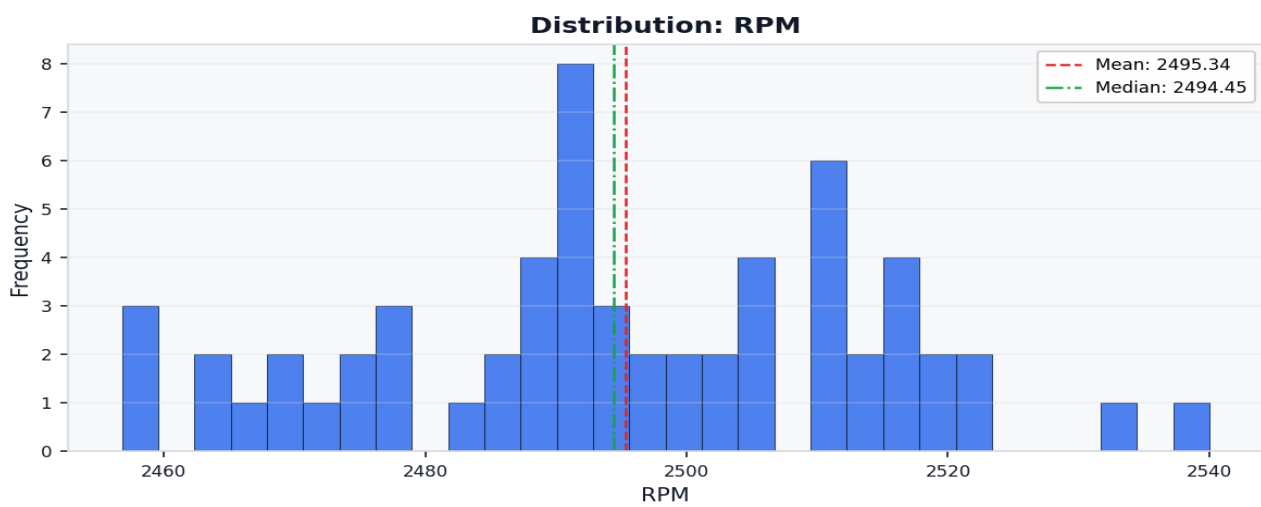
Histogram Engine Temp C



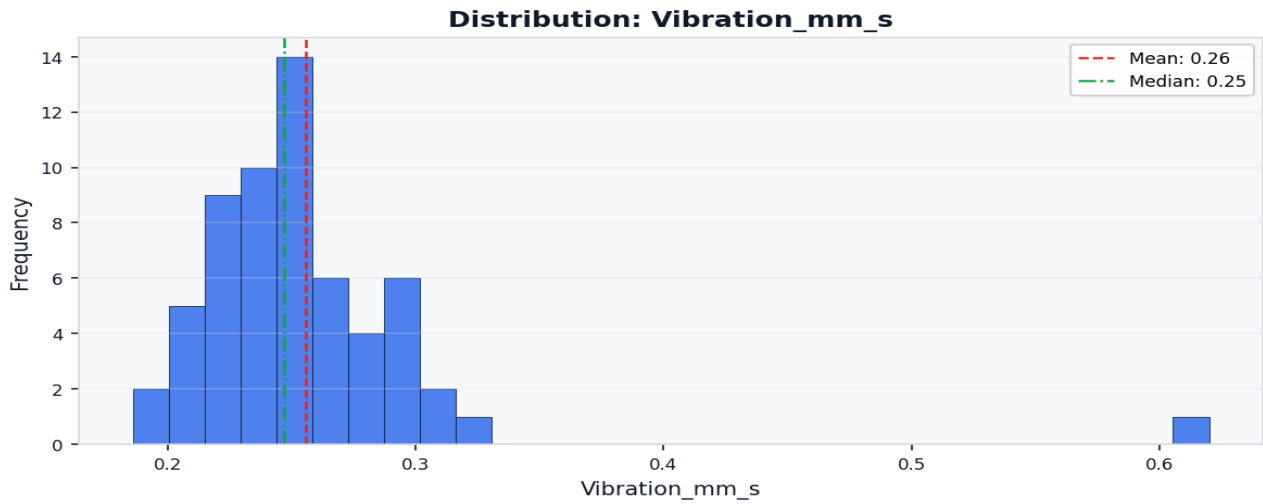
Histogram Oil Pressure Psi



Histogram Rpm



Histogram Vibration Mm S



Correlation Heatmap

